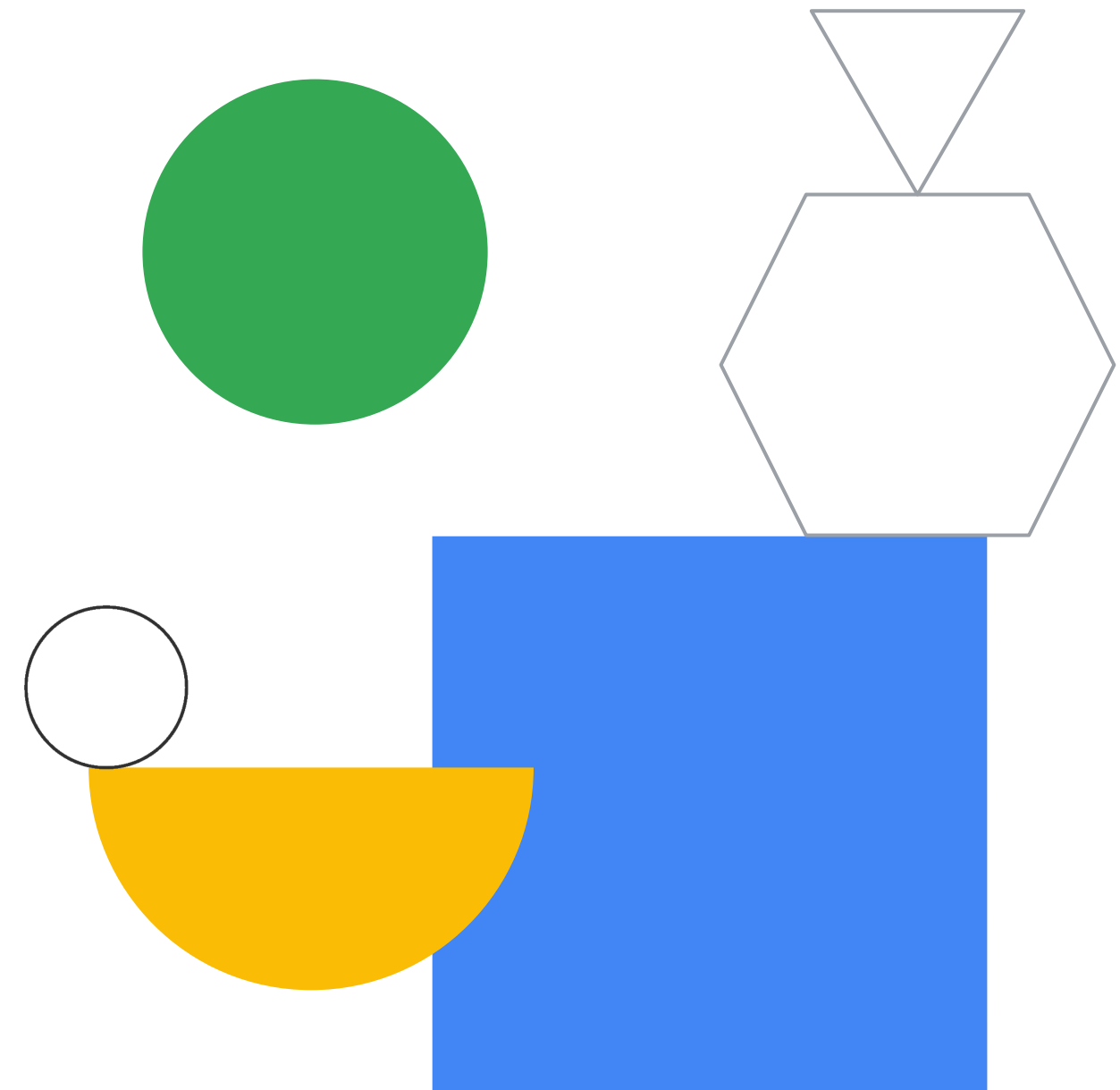
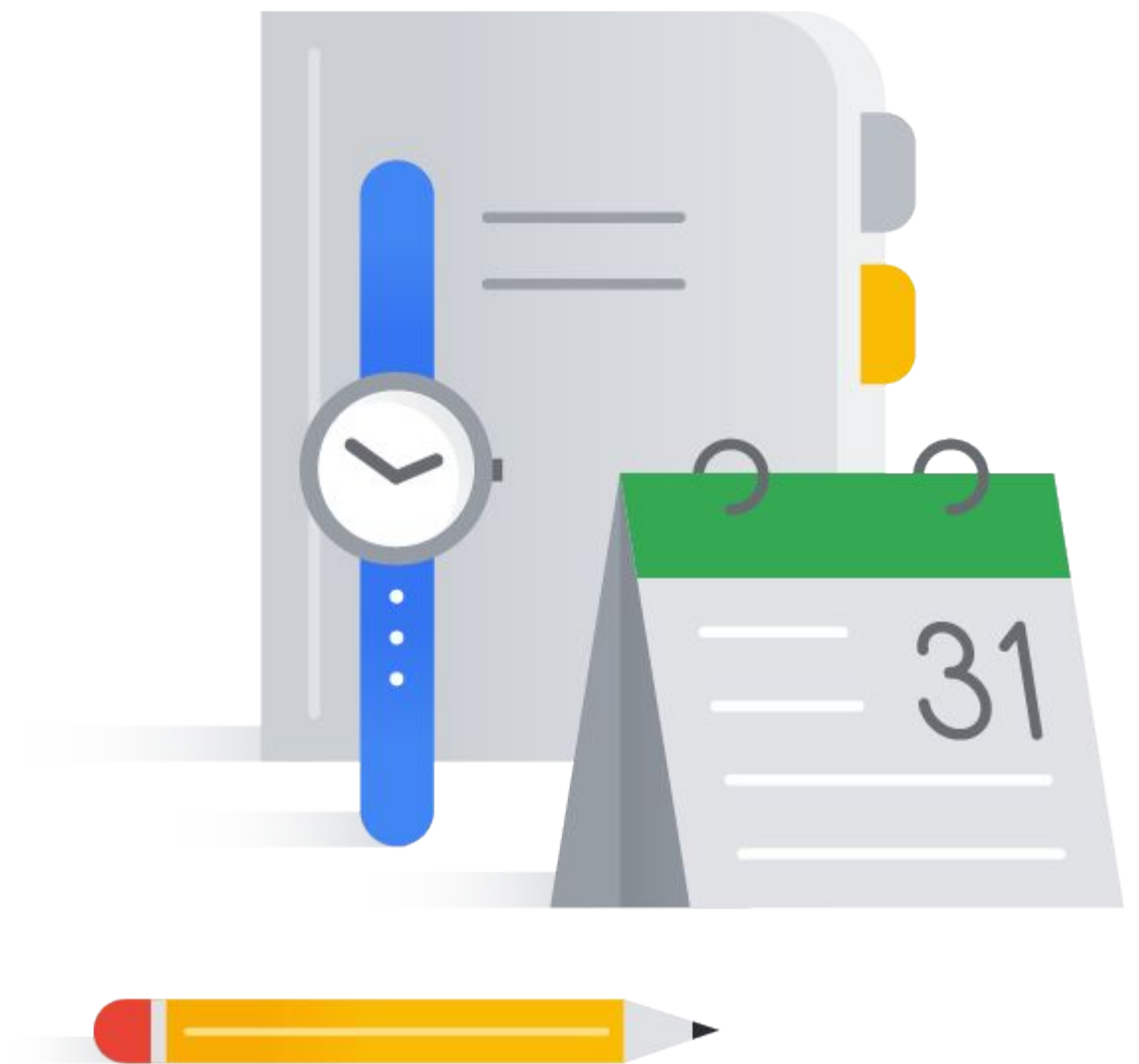


# Virtual Machines



# Agenda

- 01 Compute Engine  
Lab: Creating Virtual Machines
- 02 Compute Options (vCPU and Memory)
- 03 Images
- 04 Disk Options
- 05 Common Compute Engine Actions  
Lab: Working with Virtual Machines





# Compute Engine

# Google Cloud compute and processing options

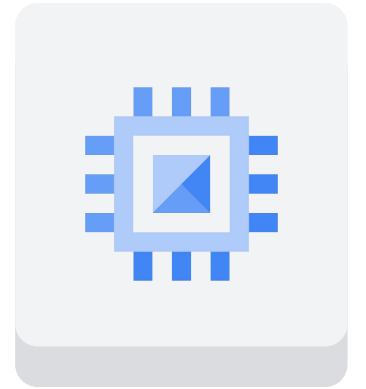
|                     | Compute Engine        | GKE                    | App Engine<br>standard<br>environment                    | App Engine<br>flexible<br>environment                                     | Cloud Run                               | Cloud Run<br>functions                                       |
|---------------------|-----------------------|------------------------|----------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------|
| Language<br>support | Any                   | Any                    | Python<br>Node.js<br>Go<br>Java<br>Ruby<br>PHP           | Python<br>Node.js<br>Go<br>Java<br>PHP<br>Ruby<br>.NET<br>Custom runtimes | Any                                     | Python<br>Node.js<br>Go<br>Java<br>.NET Core<br>Ruby<br>Java |
| Usage<br>model      | IaaS                  | IaaS<br>PaaS           | PaaS                                                     | PaaS                                                                      | PaaS                                    | Microservices<br>architecture                                |
| Scaling             | Server<br>autoscaling | Cluster                | Autoscaling managed servers                              |                                                                           | Serverless                              | Serverless                                                   |
| Primary<br>use case | General<br>workloads  | Container<br>workloads | Scalable web applications<br>Mobile backend applications |                                                                           | Deploy & scale<br>containerized<br>apps | Lightweight event<br>actions                                 |

# Compute Engine

Infrastructure as a Service (IaaS)

Predefined or custom machine types:

- vCPUs (cores) and Memory (RAM)
- Storage
  - Zonal or regional persistent disk (HDD or SSD)
  - Local SSD
  - Cloud Storage
- Networking
- Linux or Windows



Compute Engine



# Compute Engine features

## Machine rightsizing

- Recommendation engine for optimum machine size
- Cloud Monitoring statistics
- New recommendation 24 hrs after VM create or resize

- Instance metadata
- Startup and shutdown scripts

## Availability policies:

- Live migrate
- Auto restart

## Global load balancing:

- Multiple regions for availability

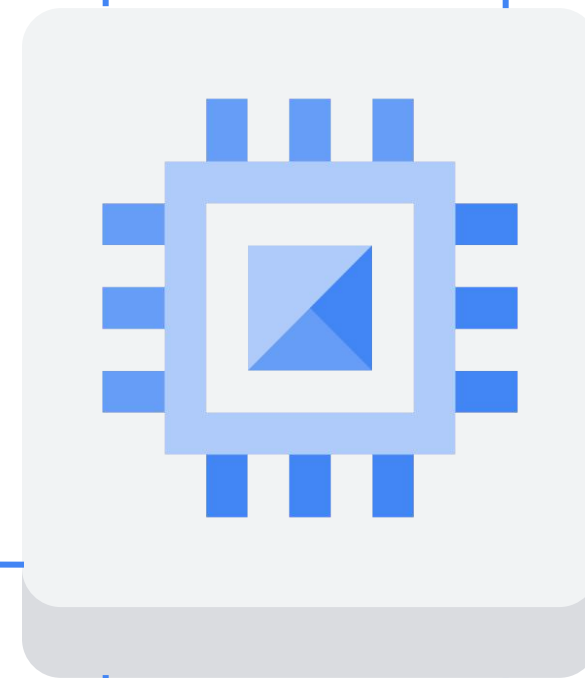
## OS patch management:

- Create patch approvals
- Set up flexible scheduling
- Apply advanced patch configuration settings

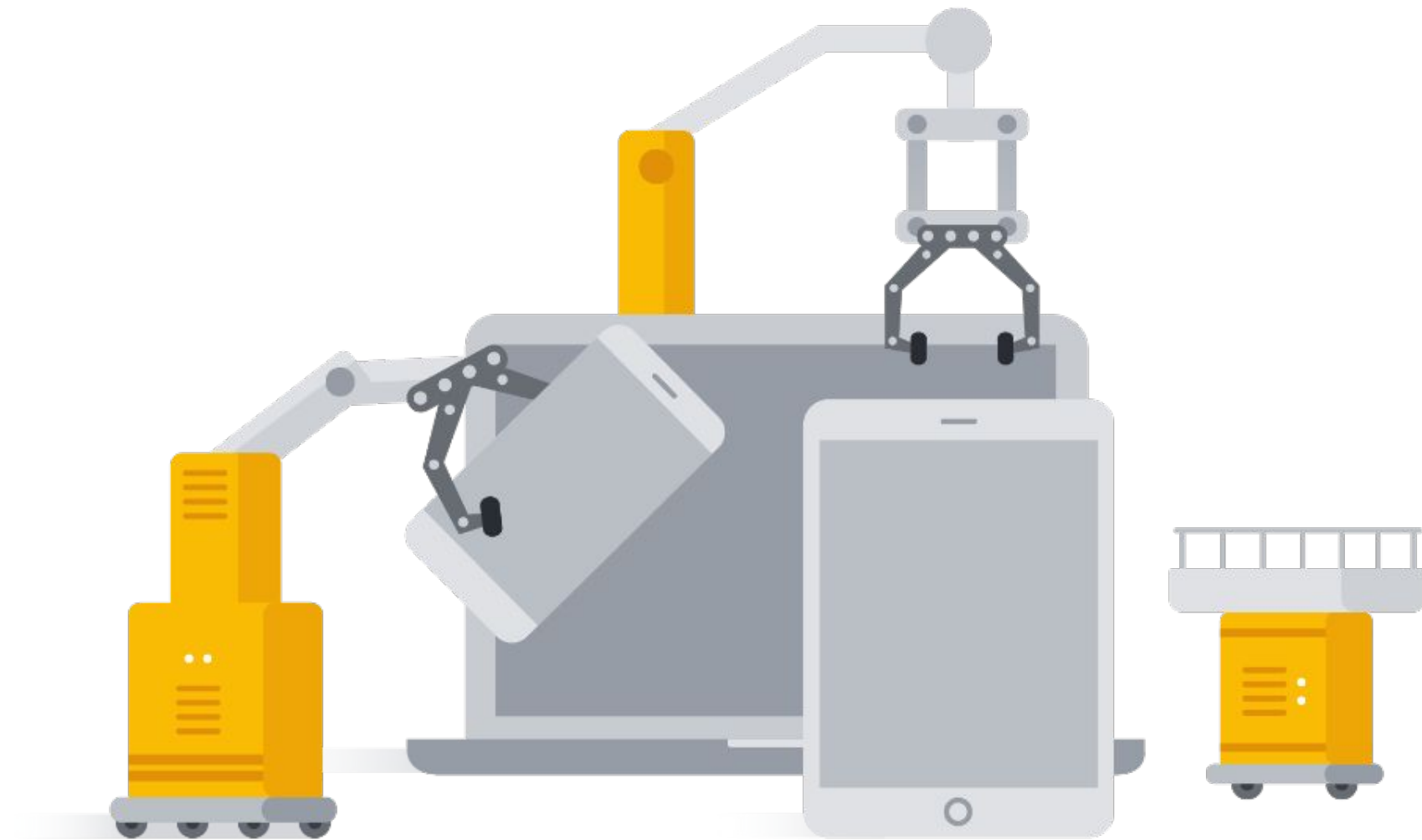
- Per-second billing
- Sustained use discounts
- Committed use discounts

## Preemptible and Spot VMs:

- Up to 91% discount
- No SLA



# Hardware limitations



**CPU**

Central processing unit



**GPU**

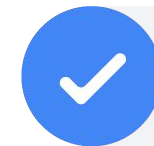
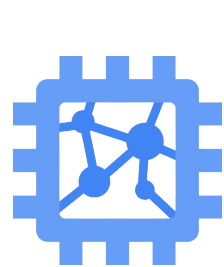
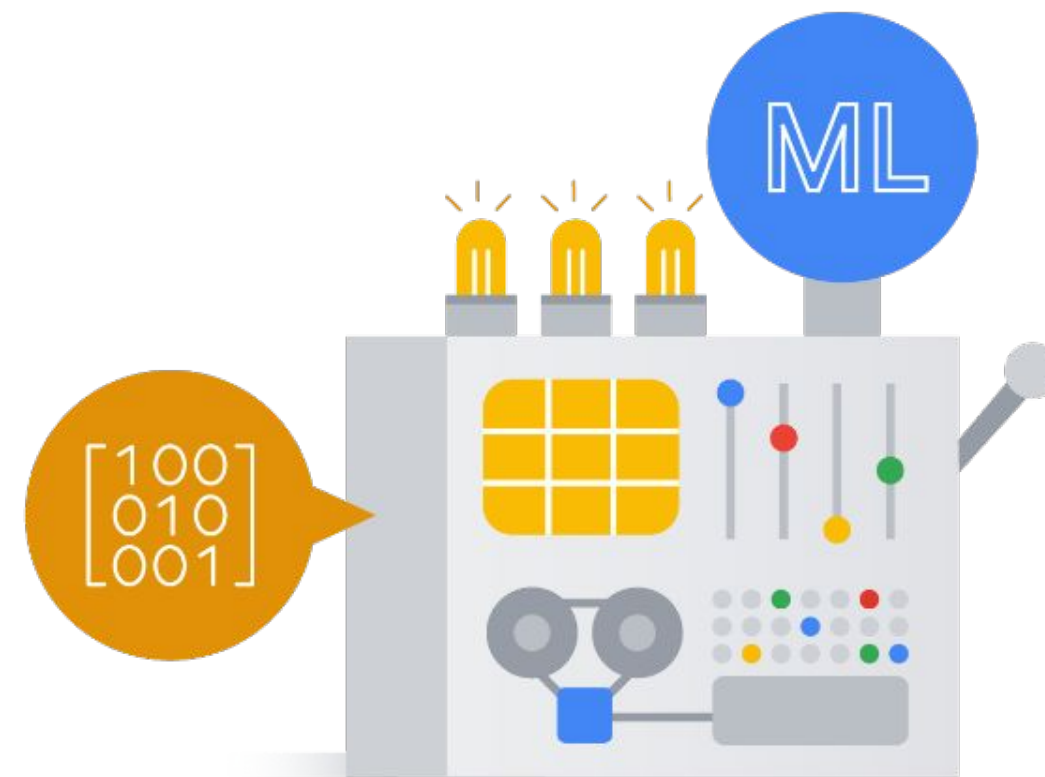
Graphics processing unit

# Tensor Processing Unit (TPU)

Domain-specific hardware

TPU

Tensor processing unit



Allow for higher efficiency



# Compute

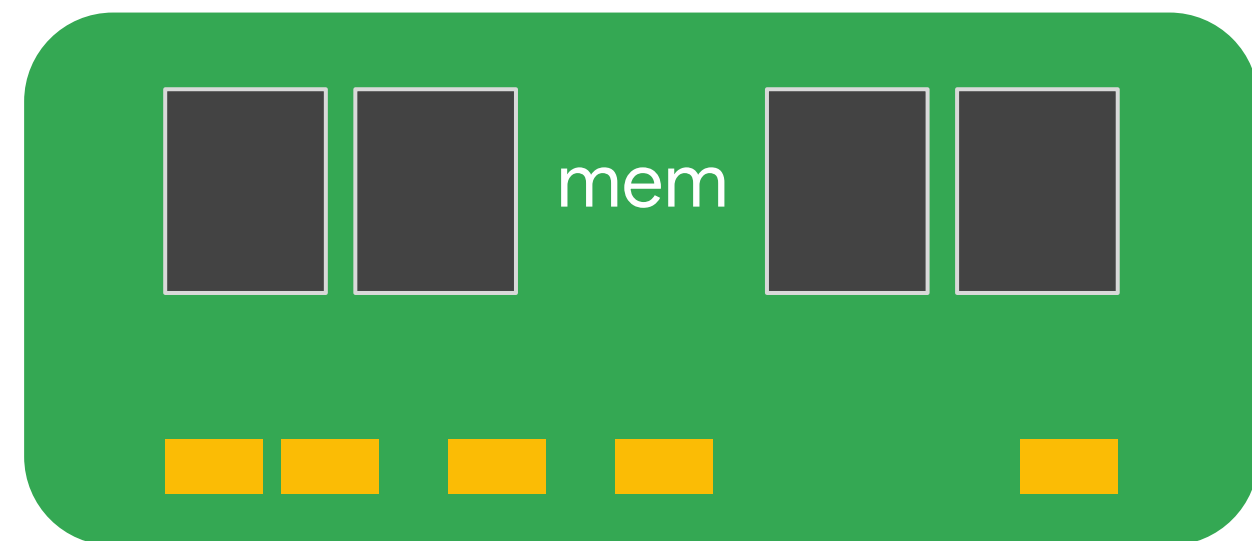
Several machine types

- Network throughput scales 2 Gbps per vCPU (small exceptions)
- Theoretical max of 200 Gbps with 176 vCPU

A vCPU is equal to 1 hardware hyper-thread



**vCPU**



# Storage

## Disks

- Standard, SSD, or Local SSD
- Standard and SSD PDs scale in performance for each GB of space allocated

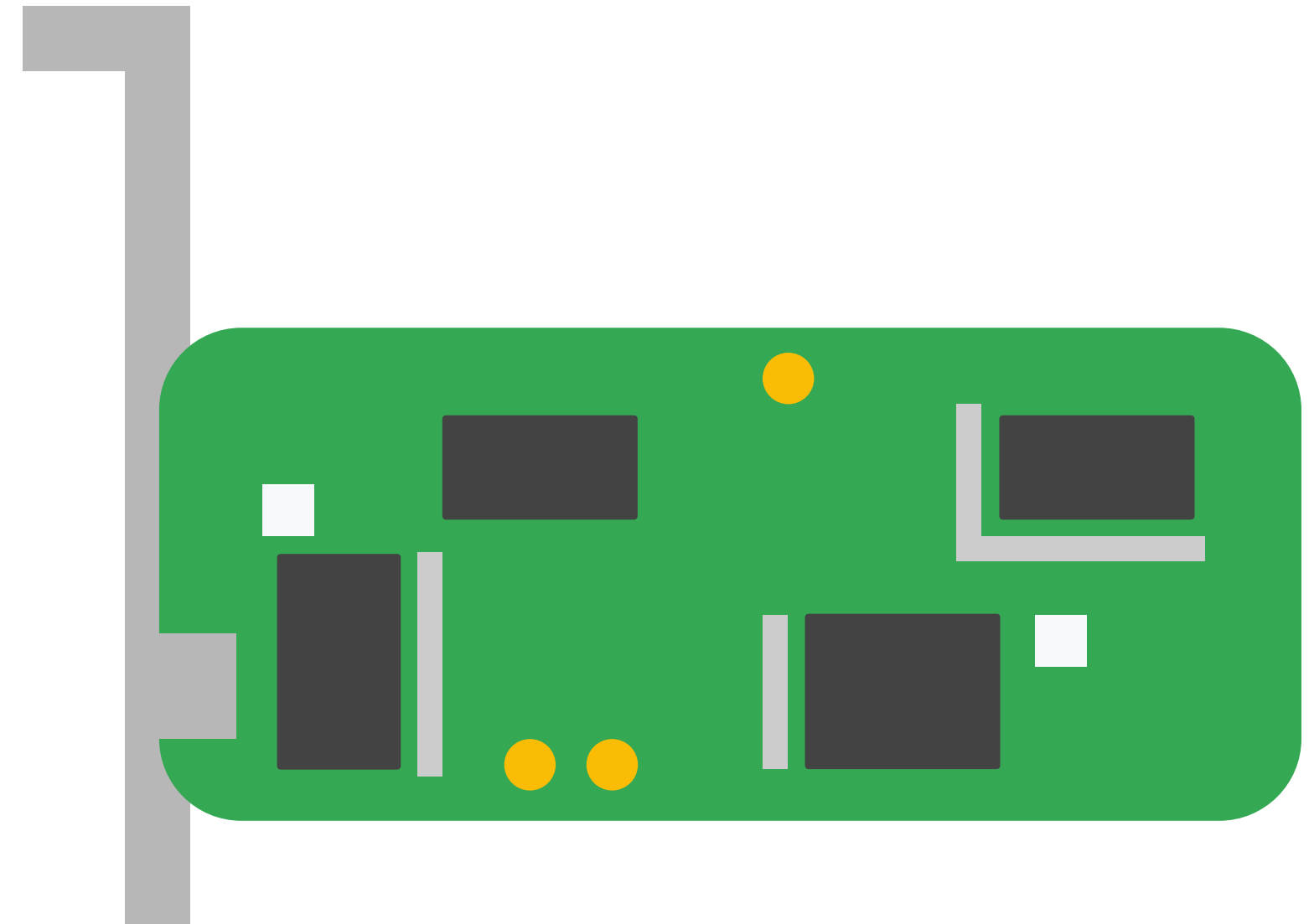
Resize disks or migrate instances with no downtime



# Networking

## Robust networking features

- Auto, custom networks
- Inbound/outbound firewall rules
  - IP based
  - Instance/group tags
- Application Load Balancing
- Network Load Balancing
  - Does not require pre-warming
- Global and multi-regional subnetworks



# VM access

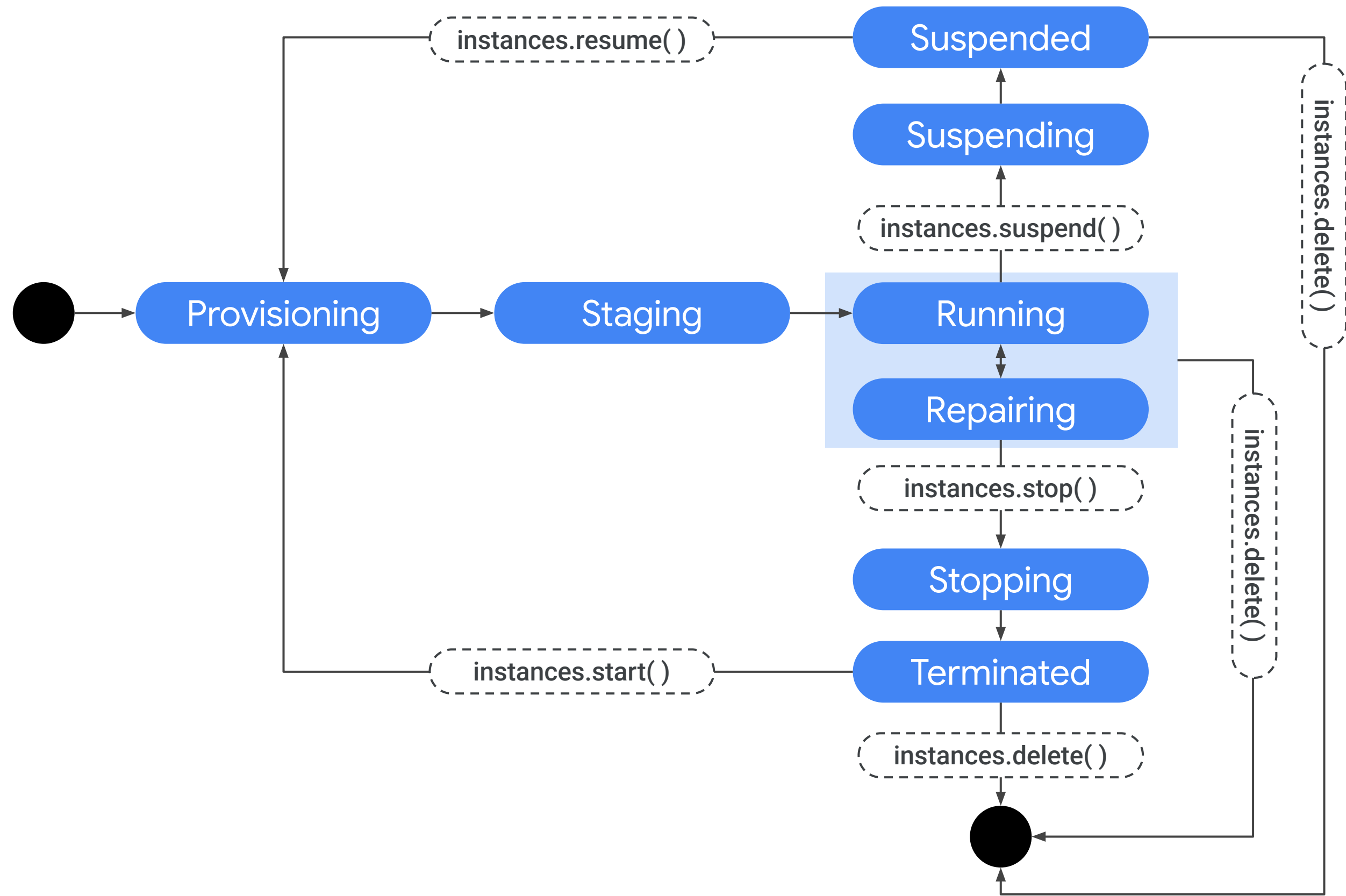
## Linux: SSH

- SSH from the Google Cloud console or Cloud Shell via the Google Cloud SDK
- SSH from computer or third-party client and generate key pair
- Requires firewall rule to allow tcp:22

## Windows: RDP

- RDP clients
- Powershell terminal
- Requires setting the Windows password
- Requires firewall rule to allow tcp:3389

# VM lifecycle





# Changing VM state from running

|                   | Methods                  | Shutdown Script time | State              |
|-------------------|--------------------------|----------------------|--------------------|
| reset             | console, gcloud, API, OS | no                   | remains running    |
| start             | console, gcloud, API     | no                   | terminated➡running |
| reboot            | OS: sudo reboot          | ~90 sec              | running➡running    |
| stop              | console, gcloud, API     | ~90 sec              | running➡terminated |
| shutdown          | OS: sudo shutdown        | ~90 sec              | running➡terminated |
| delete            | console, gcloud, API     | ~90 sec              | running➡N/A        |
| <i>preemption</i> | <i>automatic</i>         | ~30 sec              | N/A                |

"ACPI Power Off"

# Availability policy: Automatic changes

Called "scheduling options" in SDK/API

## Automatic restart

- Automatic VM restart due to crash or maintenance event
  - Not preemption or a user-initiated terminate

## On host maintenance

- Determines whether host is live-migrated or terminated due to a maintenance event.  
Live migration is the default.

## Live migration

- During maintenance event, VM is migrated to different hardware without interruption.
- Metadata indicates occurrence of live migration.

# Patch management is an essential part of managing an infrastructure

Manage OSes easily through Google Cloud.

- Keep infrastructures up-to-date.
- Reduce the risk of security vulnerabilities.

OS patch management:

- Patch compliance reporting
- Patch deployment





# There are several tasks that can be performed with patch management

Create patch approvals.

Set up flexible scheduling.

Apply advanced patch configuration settings.

Manage these patch jobs or updates from a centralized location.



# Charges for stopped (terminated) VMs

✗ Memory

✗ CPU resources



✓ Attached disks

✓ Reserved IP addresses

# Actions supported on a terminated VM



Change the machine type



Migrate the VM instance to another network



Add or remove attached disks; change auto-delete settings



Modify instance tags



Modify custom VM or project-wide metadata



Remove or set a new static IP



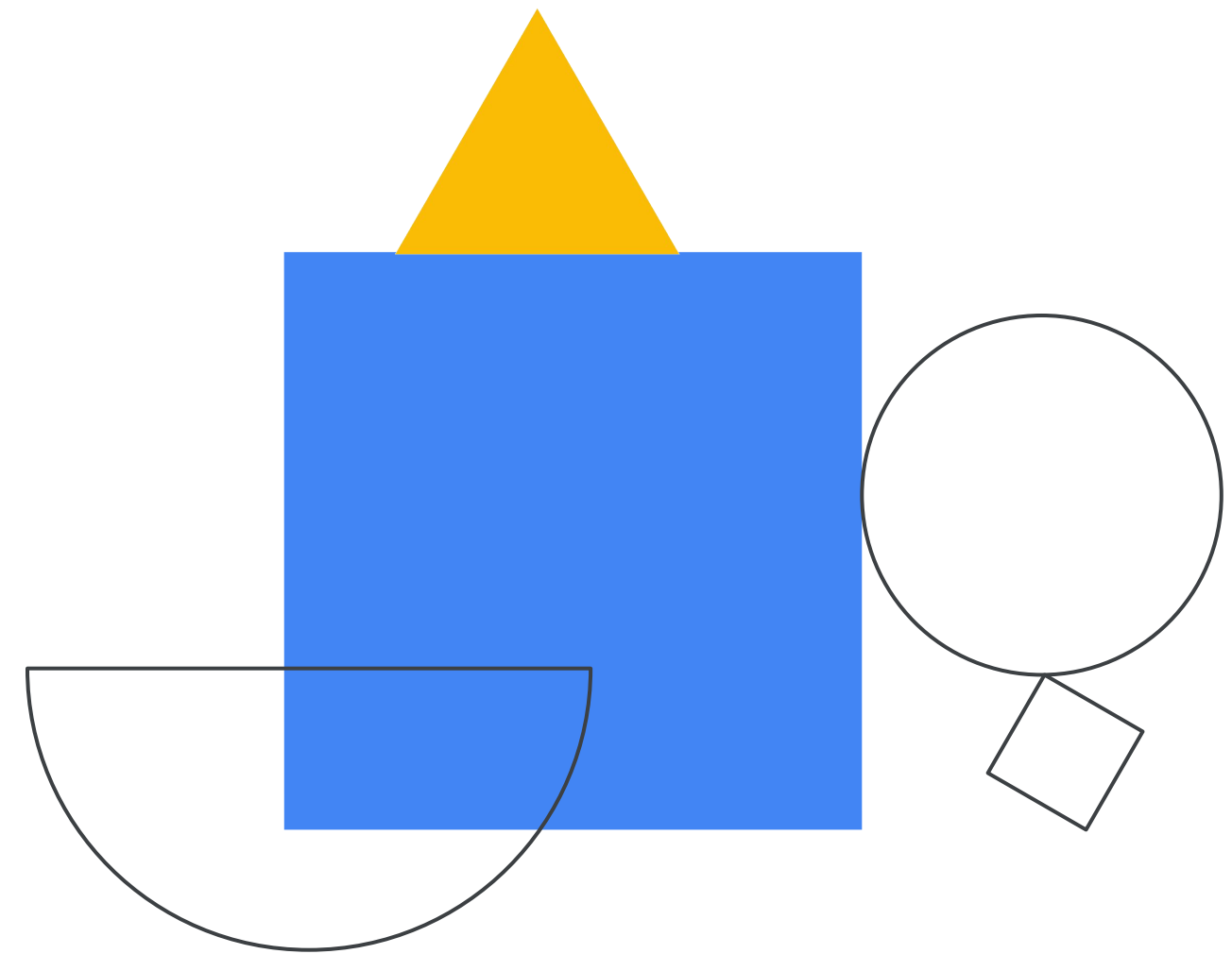
Modify VM availability policy



Can't change the image of a stopped VM

# Lab Intro

Creating Virtual Machines



# Lab objectives

01

Create several standard VMs

02

Create advanced VMs



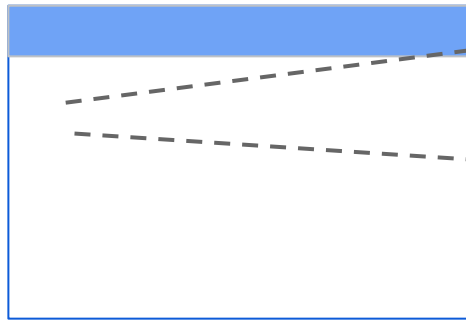




# Compute Options (vCPU and Memory)

# Creating a VM

01



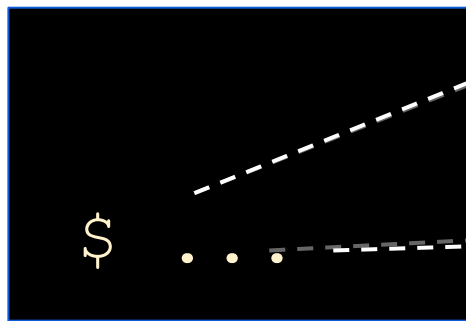
console.google.com

VM instances

CREATE INSTANCE

IMPORT VM

02



command line  
*including Cloudshell*

```
$ gcloud compute instances create [instance-name]
```

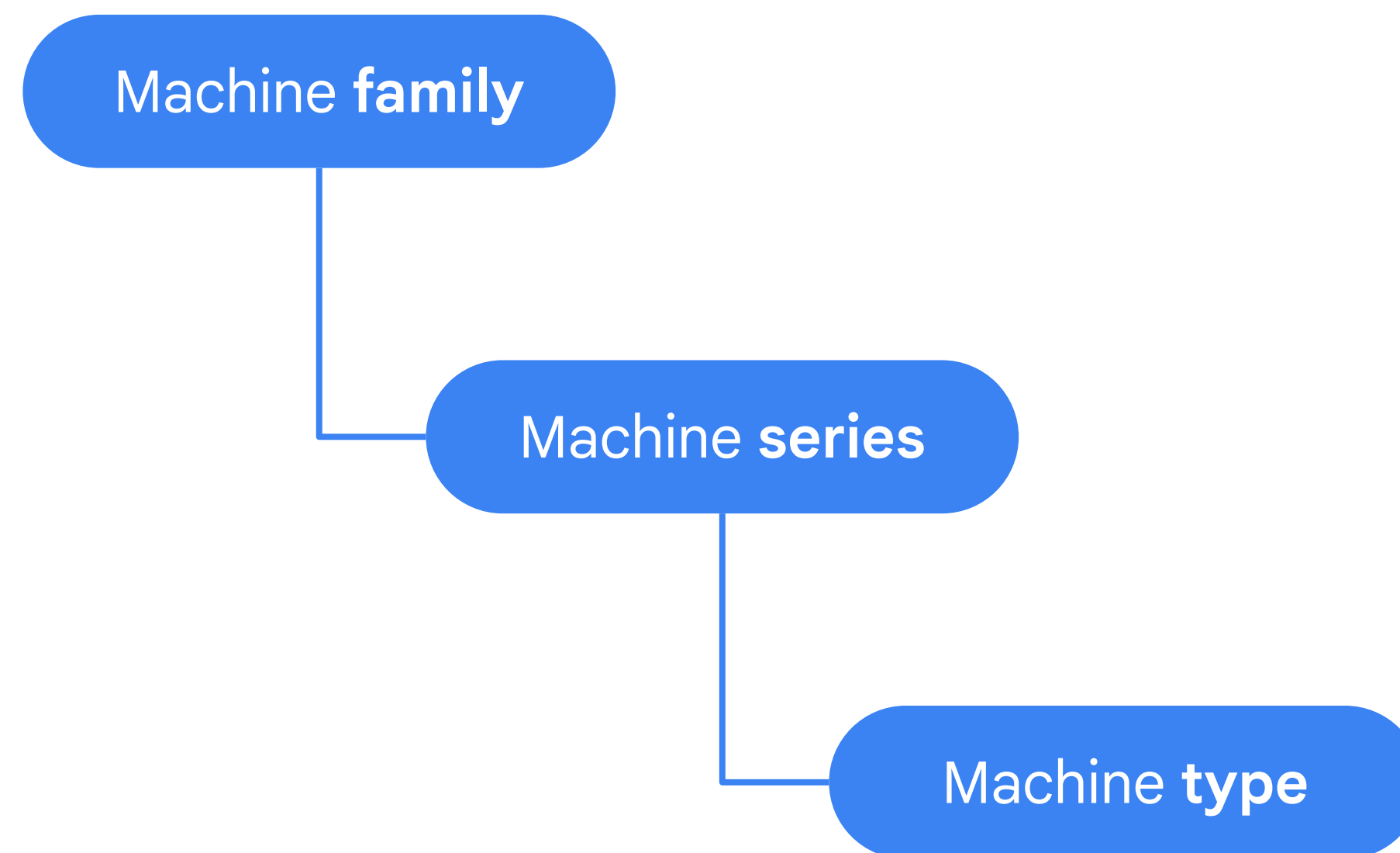
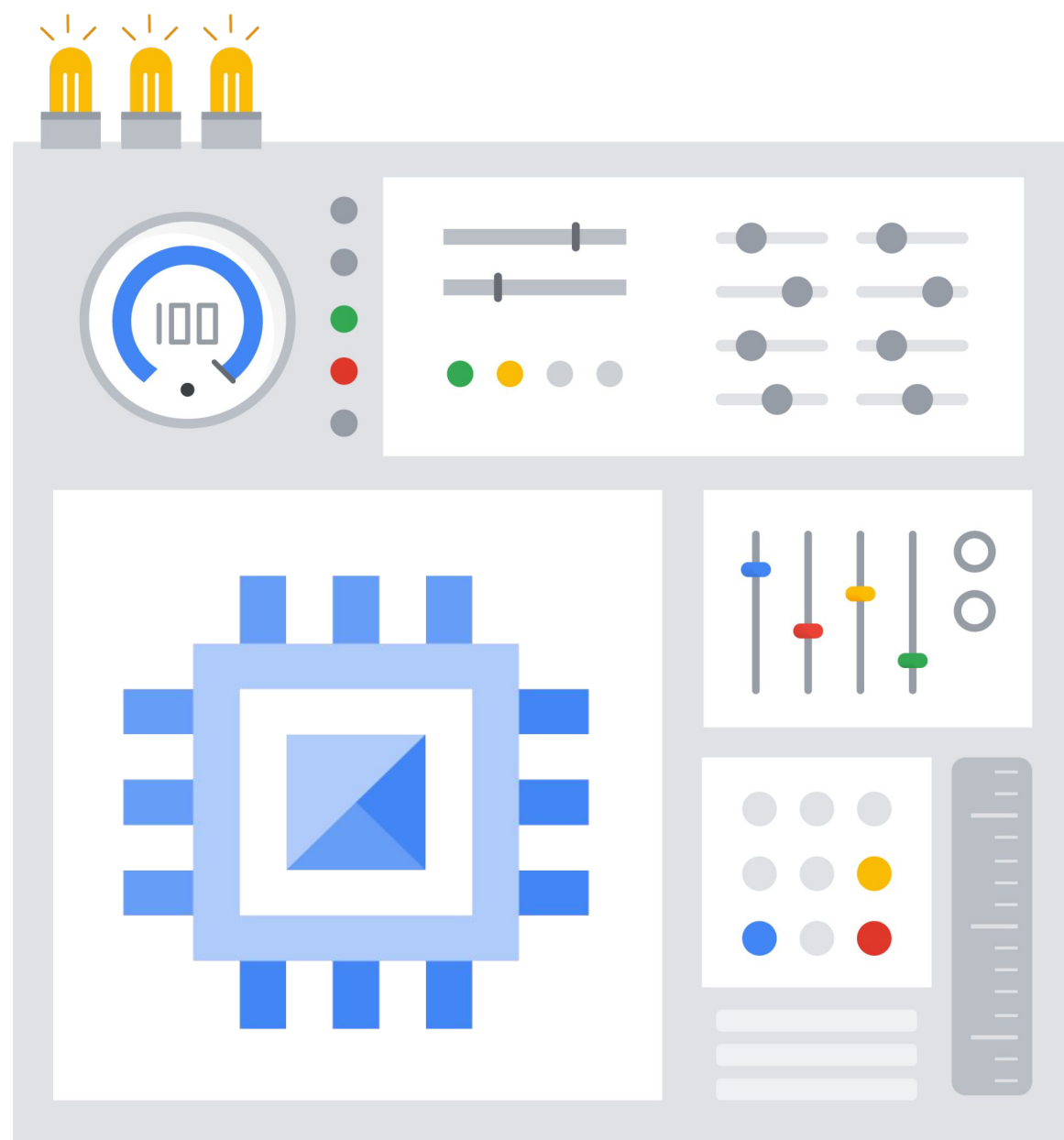
03

REST API

Many VM options

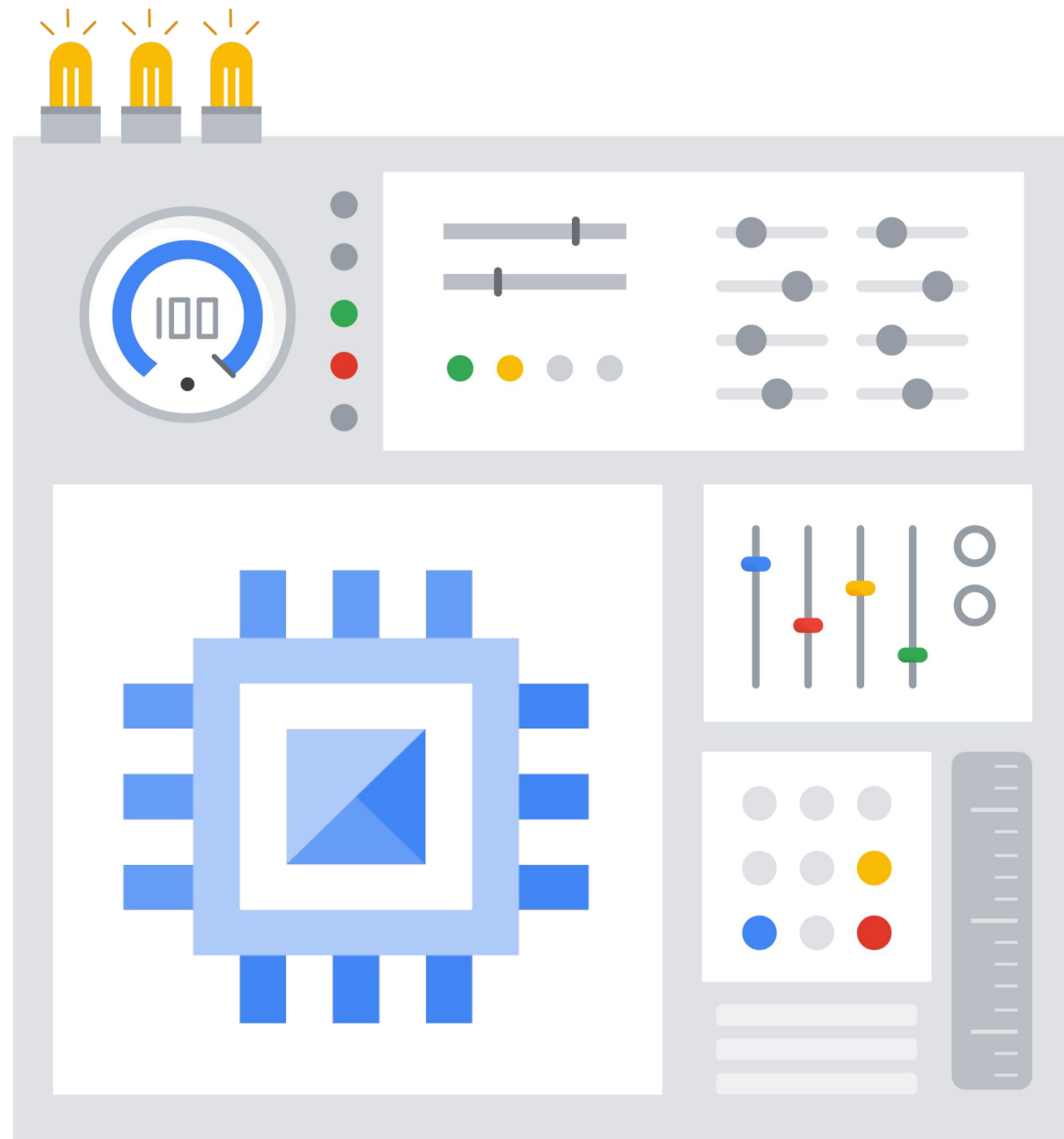
- Project
- Region
- Zone
- Subnetwork
- Machine type
- Disk options
- Image
- IP options

# Machine type structure





# Compute Engine machine families



General-purpose



Compute-optimized



Memory-optimized



Accelerator-optimized

# General-purpose machine family

| Series           | Workload                                                               | Applications                                                                                                                                    |                                                                                                                               |
|------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| E2               | (Cost-optimized) Day-to-day computing at a lower cost                  | <ul style="list-style-type: none"><li>• Web serving</li><li>• App serving</li><li>• Back office apps</li><li>• Small-medium databases</li></ul> | <ul style="list-style-type: none"><li>• Microservices</li><li>• Virtual desktops</li><li>• Development environments</li></ul> |
| N2, N2D, N1      | (Balanced) Balanced price/performance across a wide range of VM shapes | <ul style="list-style-type: none"><li>• Web serving</li><li>• App serving</li><li>• Back office apps</li></ul>                                  | <ul style="list-style-type: none"><li>• Medium-large databases</li><li>• Cache</li><li>• Media/streaming</li></ul>            |
| Tau T2D, Tau T2A | (Scale-out optimized) Best performance/ cost for scale-out workloads   | <ul style="list-style-type: none"><li>• Scale-out workloads</li><li>• Web serving</li><li>• Containerized microservices</li></ul>               | <ul style="list-style-type: none"><li>• Media transcoding</li><li>• Large-scale Java applications</li></ul>                   |

# Compute-optimized machine family

| Series | Workload                                               | Applications                                                                                                                                                              |                                                                                                                                                       |
|--------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| C2     | Ultra high performance for compute-intensive workloads | <ul style="list-style-type: none"><li>• Compute-bound workloads</li><li>• High-performance web serving</li><li>• Gaming (AAA game servers)</li><li>• Ad serving</li></ul> | <ul style="list-style-type: none"><li>• High-performance computing (HPC)</li><li>• Media transcoding</li><li>• AI/ML</li></ul>                        |
| C2D    | Ultra high performance for compute-intensive workloads | <ul style="list-style-type: none"><li>• Memory-bound workloads</li><li>• Gaming (AAA game servers)</li><li>• High-performance computing (HPC)</li></ul>                   | <ul style="list-style-type: none"><li>• High-performance databases</li><li>• Electronic Design Automation (EDA)</li><li>• Media transcoding</li></ul> |
| H3     | Ultra high performance for compute-intensive workloads | <ul style="list-style-type: none"><li>• High-performance computing (HPC)</li></ul>                                                                                        | <ul style="list-style-type: none"><li>• Electronic Design Automation (EDA)</li></ul>                                                                  |

# Memory-optimized machine family

| Series | Workload                    | Applications                                                                                                                                                                                                                             |                                                                                                                                                                                                                                     |
|--------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1     | Ultra high-memory workloads | <ul style="list-style-type: none"><li>• Medium in-memory databases such as SAP HANA</li><li>• Tasks that require intensive use of memory with higher memory-to-vCPU ratios than the general-purpose high-memory machine types.</li></ul> | <ul style="list-style-type: none"><li>• In-memory databases and in-memory analytics, business warehousing (BW) workloads, genomics analysis, SQL analysis services.</li><li>• Microsoft SQL Server and similar databases.</li></ul> |
| M2     | Ultra high-memory workloads | <ul style="list-style-type: none"><li>• Large in-memory databases such as SAP HANA</li></ul>                                                                                                                                             | <ul style="list-style-type: none"><li>• In-memory databases and in-memory analytics, business warehousing (BW) workloads, genomics analysis, SQL analysis services, etc.</li></ul>                                                  |
| M3     | Ultra high-memory workloads | <ul style="list-style-type: none"><li>• OLAP and OLTP SAP workloads</li></ul>                                                                                                                                                            | <ul style="list-style-type: none"><li>• Memory Intense Electronic Design Automation</li></ul>                                                                                                                                       |

# Accelerator-optimized machine family

| Series | Workload                                           | Applications                                                                                                                                      |
|--------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| A2     | Optimized for high-performance computing workloads | <ul style="list-style-type: none"><li>• CUDA-enabled ML training and inference</li><li>• HPC</li><li>• Massive parallelized computation</li></ul> |
| G2     | Optimized for high-performance computing workloads | <ul style="list-style-type: none"><li>• Video transcoding</li><li>• Remote visualization workstation</li></ul>                                    |



# Creating custom machine types

## When to select custom:

- Requirements fit between the predefined types
- Need more memory or more CPU

## Customize the amount of memory and vCPU for your machine:

- Either 1 vCPU or even number of vCPU
- Up to 8 GB per vCPU
- Total memory must be multiple of 256 MB

### Machine configuration

✓ General purpose Compute optimized Memory optimized GPUs

Machine types for common workloads, optimized for cost and flexibility

Series

N2

Powered by Intel Cascade Lake and Ice Lake CPU platforms

#### Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

PRESET CUSTOM

Creating a custom machine incurs additional costs

Cores

2

80

2 vCPU

Memory

1

16

16 GB

☐ Extend Memory ?

# Pricing

- Per-second billing, with minimum of 1 minute
  - vCPUs, GPUs, and GB of memory
- Resource-based pricing:
  - Each vCPU and each GB of memory is billed separately
- Discounts:
  - Sustained use
  - Committed use
  - Preemptible and Spot VM instances
- Recommendation Engine
  - Notifies you of underutilized instances
- Free usage limits

# Sustained use discounts

## Sustained use discounts for up to 30%

General-purpose N1 predefined and custom machine types, memory-optimized machine types, shared-core machine types, and sole-tenant nodes

| Usage level<br>(% of month) | % at which incremental<br>is charged |
|-----------------------------|--------------------------------------|
| 0% - 25%                    | 100% of base rate                    |
| 25% - 50%                   | 80% of base rate                     |
| 50% - 75%                   | 60% of base rate                     |
| 75% - 100%                  | 40% of base rate                     |

## Sustained use discounts for up to 20%

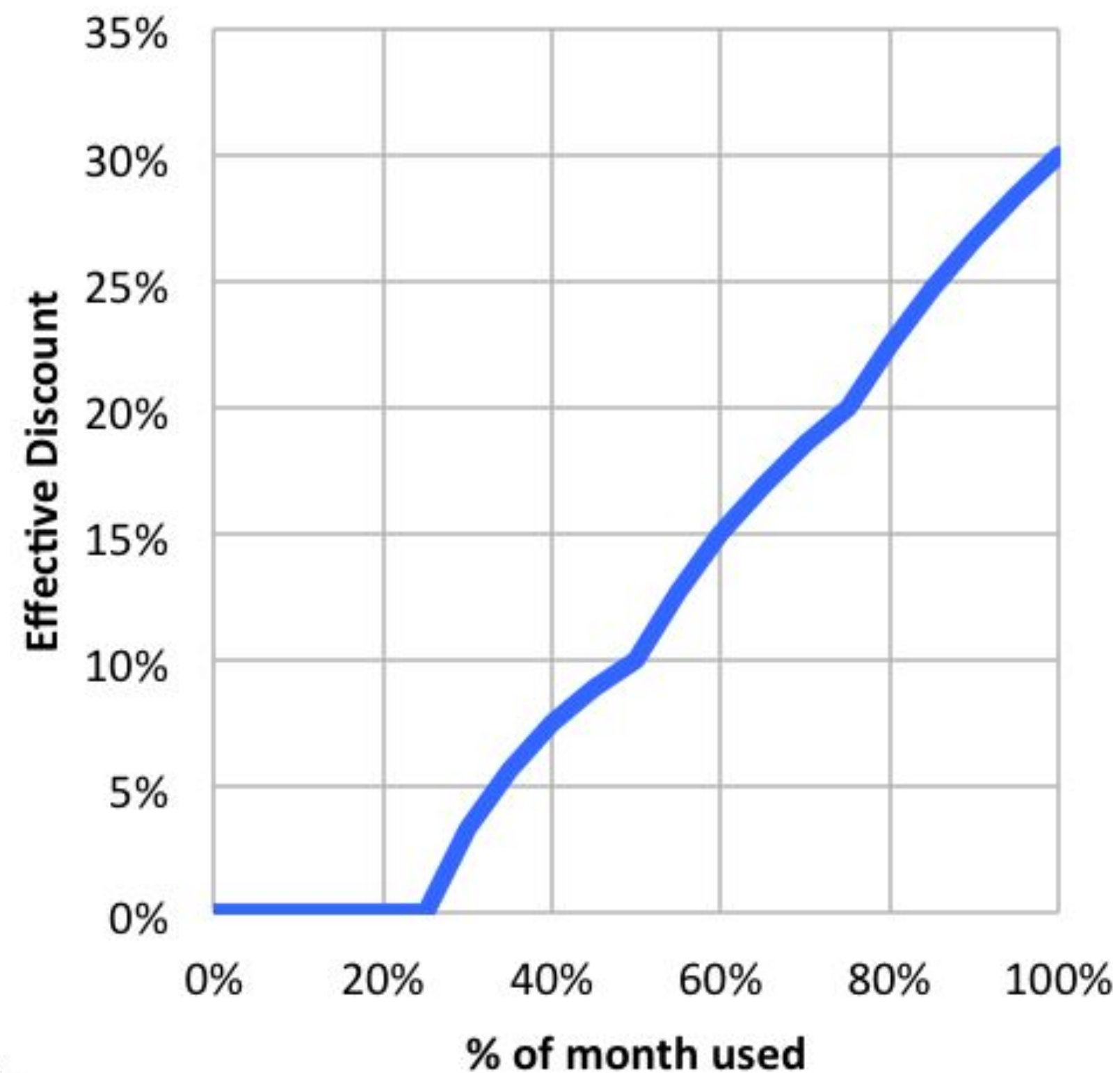
General-purpose N2 and N2D predefined and custom machine types, and compute-optimized machine types

| Usage level<br>(% of month) | % at which incremental<br>is charged |
|-----------------------------|--------------------------------------|
| 0% - 25%                    | 100% of base rate                    |
| 25% - 50%                   | 86.78% of base rate                  |
| 50% - 75%                   | 73.3% of base rate                   |
| 75% - 100%                  | 60% of base rate                     |

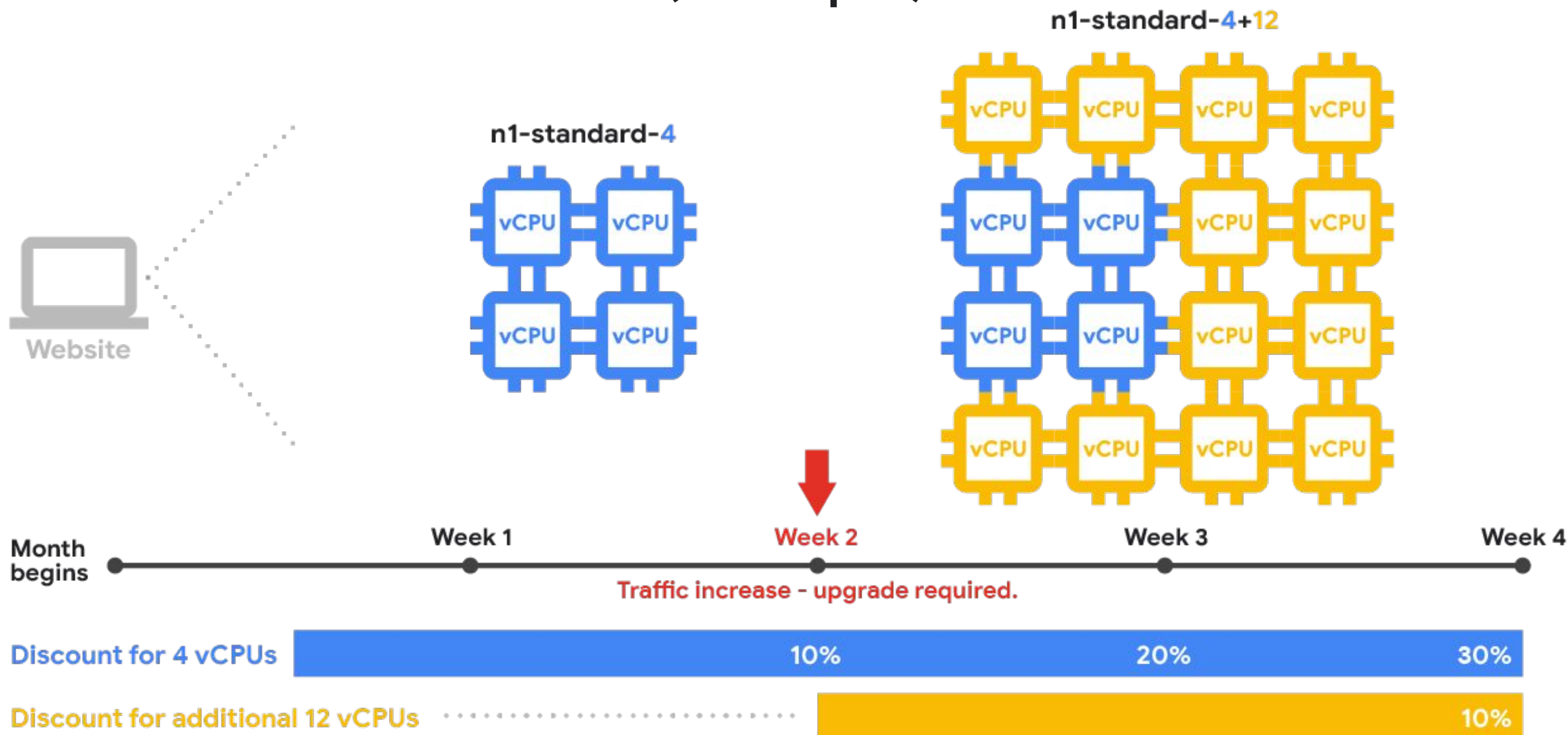
Up to 30% net discount for instances that run the entire month



Sustained use  
discounts  
increase with use



# Sustained use discounts (example)



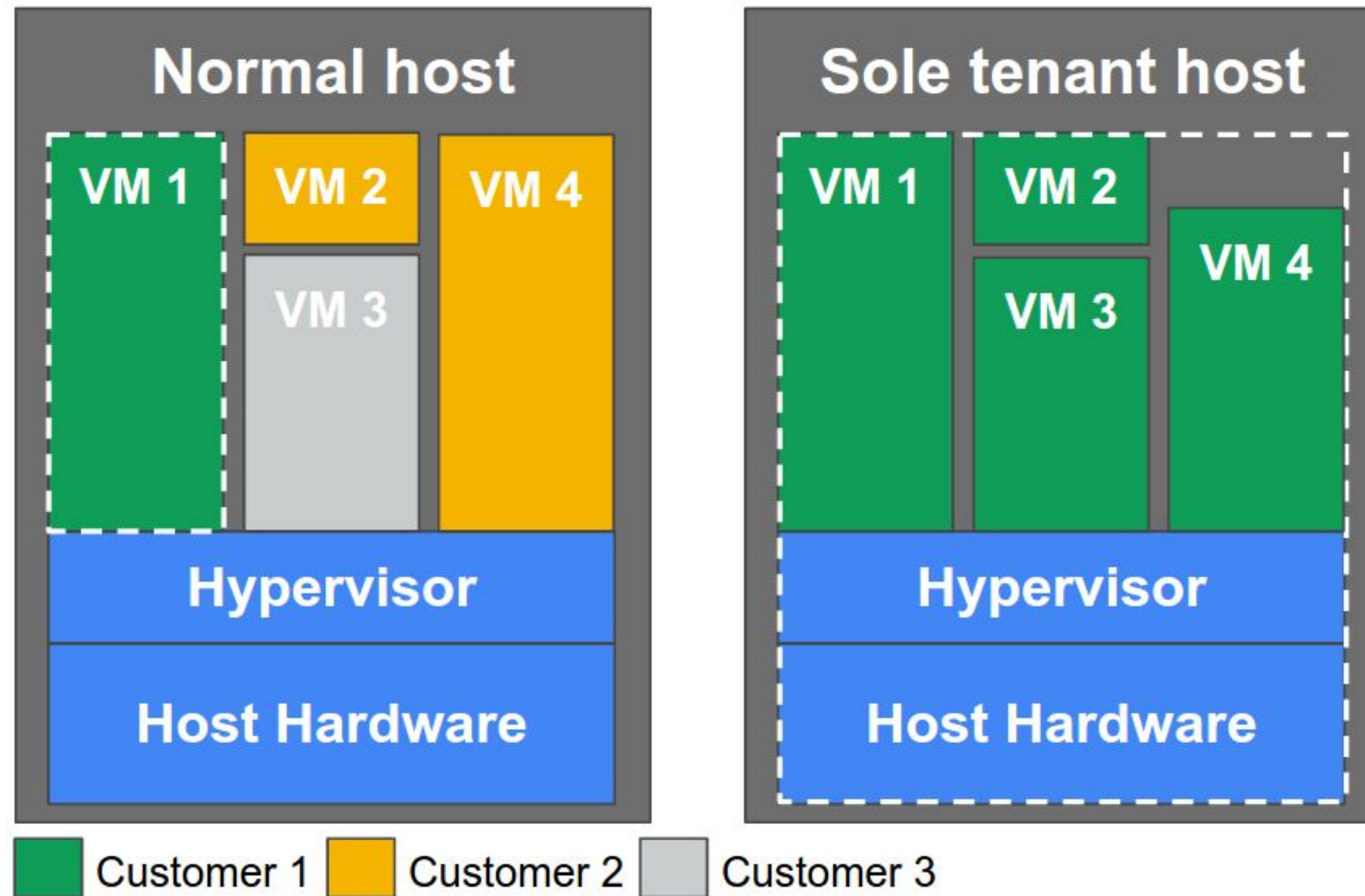
# Preemptible

- Lower price for interruptible service (up to 91%)
- VM might be terminated at any time
  - No charge if terminated in the first minute
  - 24 hours max
  - 30-second terminate warning, but not guaranteed
    - Time for a shutdown script
- No live migrate; no auto restart
- You can request that CPU quota for a region be split between regular and preemption
  - Default: preemptible VMs count against region CPU quota

# Spot VMs

- Spot VMs are the latest version of preemptible VMs
- Spot VMs and preemptible VMs share the same pricing model
- No minimum or maximum runtime
- Spot VMs are finite Compute Engine resources, so they might not always be available
  - No live migrate; no auto restart
- Best practice use cases help you get the most of using Spot VMs

# Sole-tenant nodes physically isolate workloads

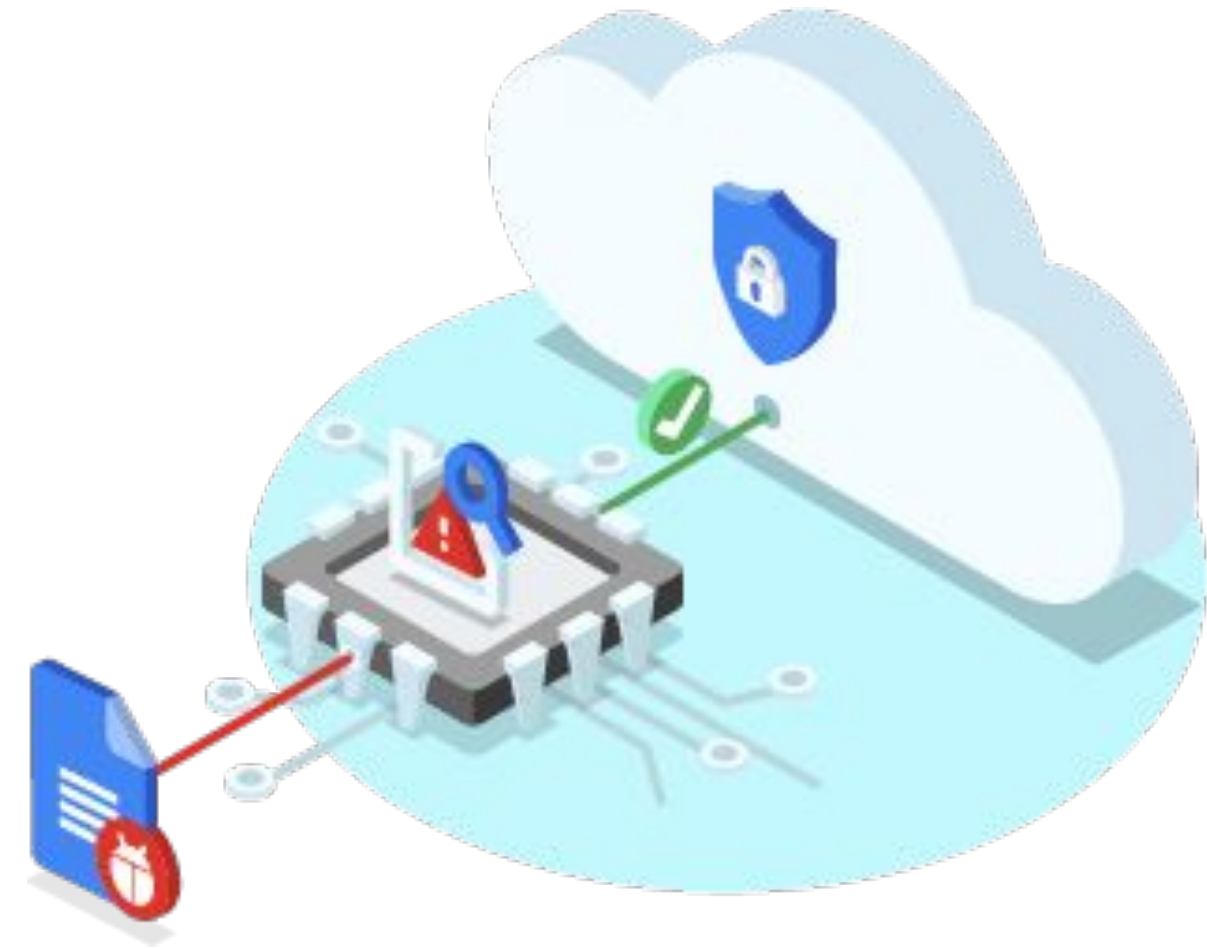


You can  
bring  
your own  
license!



# Shielded VMs offer verifiable integrity

- Secure Boot
- Virtual trusted platform module (vTPM)
- Integrity monitoring



Requires shielded image!

# Confidential VMs allow you to encrypt data in use

- Encrypts data while it's being processed.
- Easy to use with no changes to code or performance compromise.
- N2D Compute Engine VM running on second generation AMD Epyc processors.
- Provides high memory capacity, high throughput, and supports parallel and compute heavy workloads.
- You can select Confidential VM service when creating a new VM.

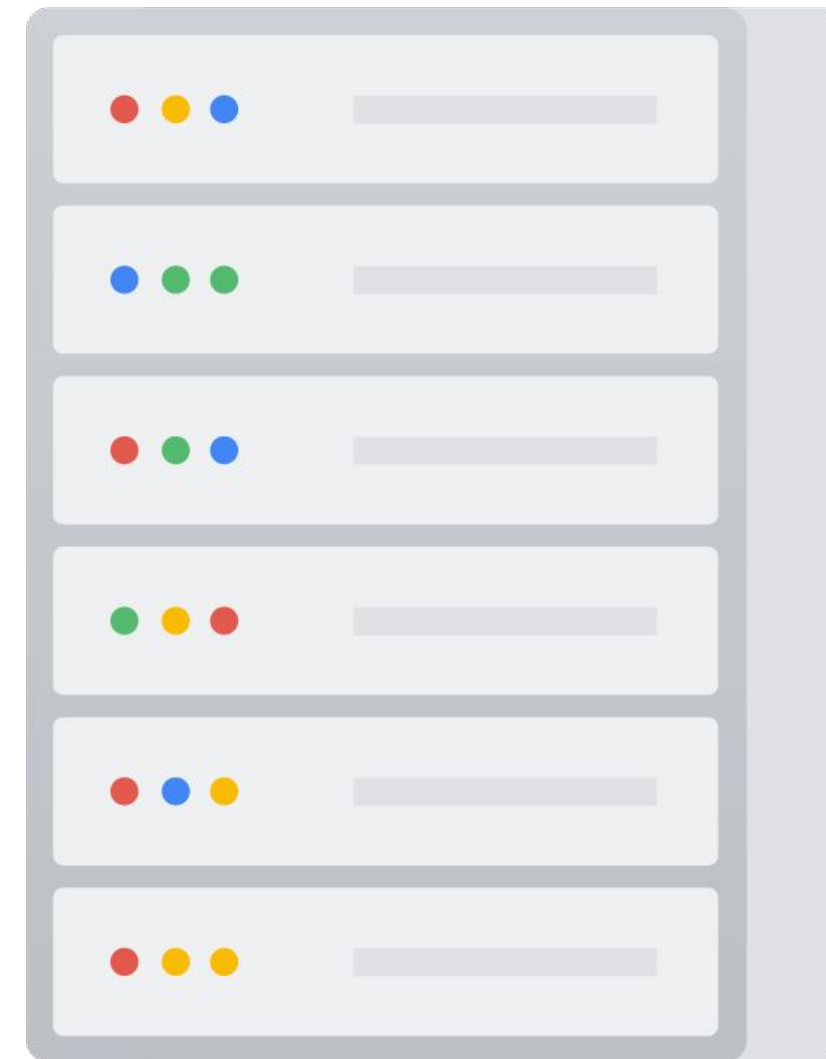


# Images



# What's in an image?

- Boot loader
- Operating system
- File system structure
- Software
- Customizations



# Images

## Public base images

- Google, third-party vendors, and community; Premium images (p)
- Linux
  - CentOS, CoreOS, Debian, RHEL(p), SUSE(p), Ubuntu, openSUSE, and FreeBSD
- Windows
  - Windows Server 2019(p), 2016(p), 2012-r2(p)
  - SQL Server pre-installed on Windows(p)

## Custom images

- Create new image from VM: pre-configured and installed SW
- Import from on-prem, workstation, or another cloud
- Management features: image sharing, image family, deprecation

# Machine images

| Scenarios              | Machine image | Persistent disk snapshot | Custom image | Instance template |
|------------------------|---------------|--------------------------|--------------|-------------------|
| Single disk backup     | Yes           | Yes                      | Yes          | No                |
| Multiple disk backup   | Yes           | No                       | No           | No                |
| Differential backup    | Yes           | Yes                      | No           | No                |
| Instance cloning       | Yes           | No                       | Yes          | Yes               |
| Base image replication | No            | No                       | Yes          | No                |



# Disk Options

# Boot disk

- VM comes with a single root persistent disk.
- Image is loaded onto root disk during first boot:
  - Bootable: you can attach to a VM and boot from it.
  - Durable: can survive VM terminate.
- Some OS images are customized for Compute Engine.
- Can survive VM deletion if “Delete boot disk when instance is deleted” is disabled.



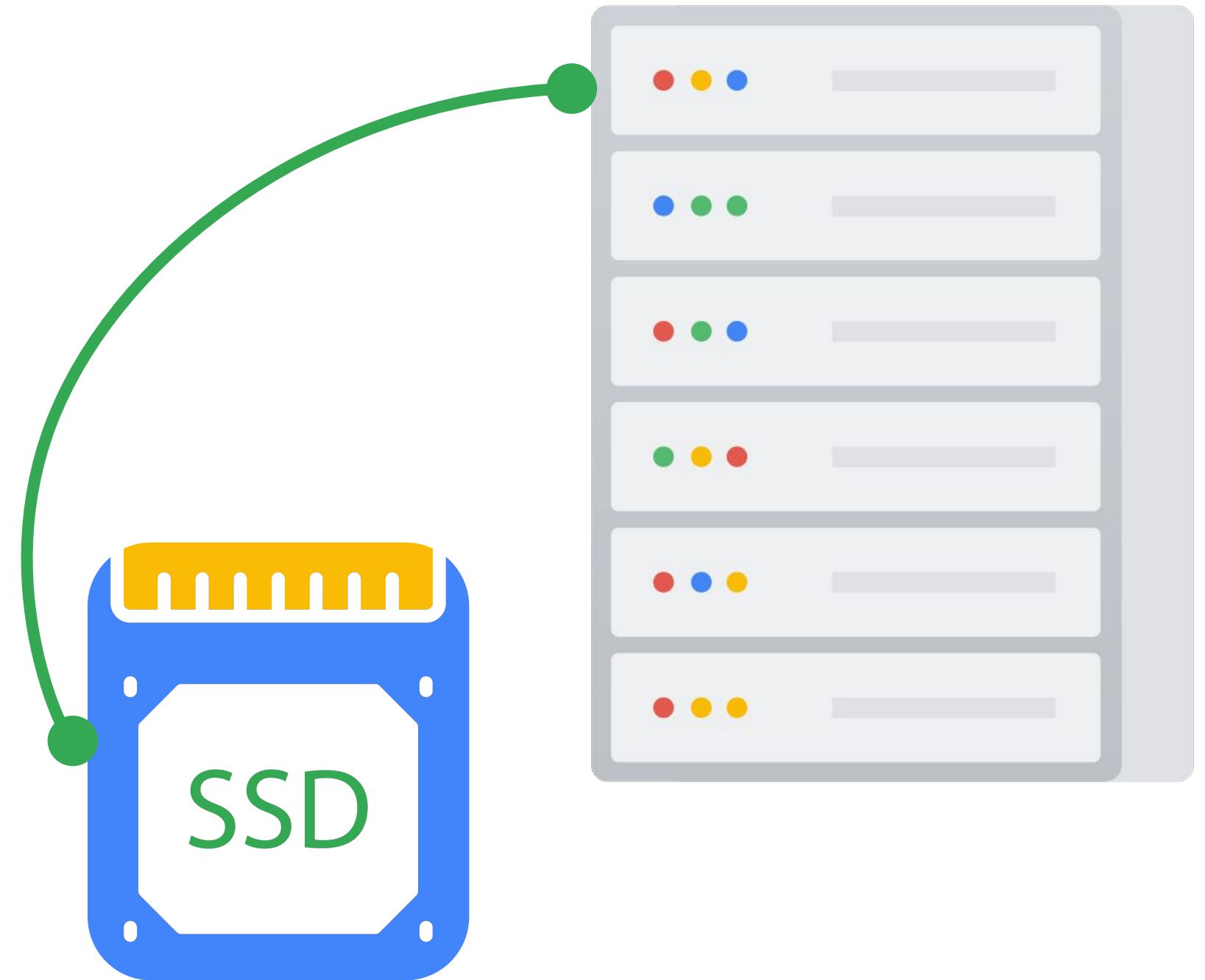
# Persistent disks

Network storage appearing as a block device

- Attached to a VM through the network interface
- Durable storage: can survive VM terminate
- Bootable: you can attach to a VM and boot from it
- Snapshots: incremental backups
- Performance: Scales with size
- HDD (magnetic) or SSD (faster, solid-state) options
- Disk resizing: even running and attached!
- Can be attached in read-only mode to multiple VMs
- Zonal or Regional
  - pd-standard
  - pd-ssd
  - pd-balanced
  - pd-extreme (zonal only)
- Encryption keys:
  - Google-managed
  - Customer-managed
  - Customer-supplied

# Local SSD disks are physically attached to a VM

- More IOPS, lower latency, and higher throughput than persistent disk
- 375-GB disk up to 24, total of 9 TB per instance
- Data survives a reset, but not a VM stop or terminate
- VM-specific: cannot be reattached to a different VM



# RAM disk

- tmpfs
- Faster than local disk, slower than memory
  - Use when your application expects a file system structure and cannot directly store its data in memory
  - Fast scratch disk, or fast cache
- Very volatile; erase on stop or restart
- May need a larger machine type if RAM was sized for the application
- Consider using a persistent disk to back up RAM disk data



# Summary of disk options

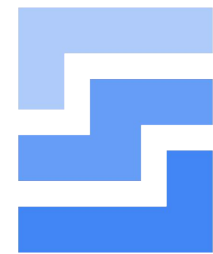
|                       | Persistent disk<br>HDD        | Persistent disk<br>SSD | Local SSD disk               | RAM disk                             |
|-----------------------|-------------------------------|------------------------|------------------------------|--------------------------------------|
| Data<br>redundancy    | Yes                           | Yes                    | No                           | No                                   |
| Encryption at<br>rest | Yes                           | Yes                    | Yes                          | N/A                                  |
| Snapshotting          | Yes                           | Yes                    | No                           | No                                   |
| Bootable              | Yes                           | Yes                    | No                           | Not                                  |
| Use case              | General, bulk file<br>storage | Very random<br>IOPS    | High IOPS and<br>low latency | low latency and<br>risk of data loss |

# Maximum persistent disks

| Machine type      | Disk number limit |
|-------------------|-------------------|
| Shared-core       | 16                |
| Standard          | 128               |
| High-memory       |                   |
| High-CPU          |                   |
| Memory-optimized  |                   |
| Compute-optimized |                   |

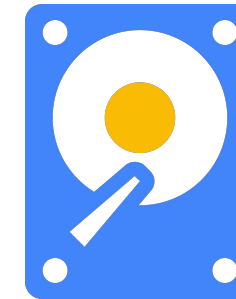
# Persistent disk management differences

## Cloud Persistent Disk



- Single file system is best
- Resize (grow) disks
- Resize file system
- Built-in snapshot service
- Automatic encryption

## Computer Hardware Disk



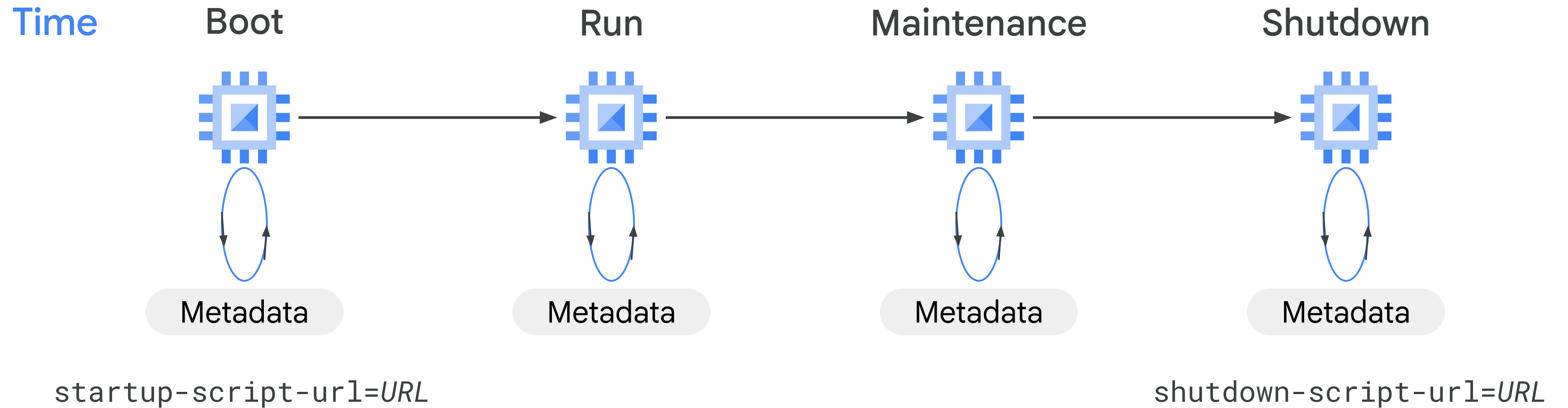
- Partitioning
- Repartition disk
- Reformat
- Redundant disk arrays
- Subvolume management and snapshots
- Encrypt files before write to disk

A blue square containing the white text '05' in a large, bold, sans-serif font. To the right of the square is a yellow downward-pointing triangle.

05

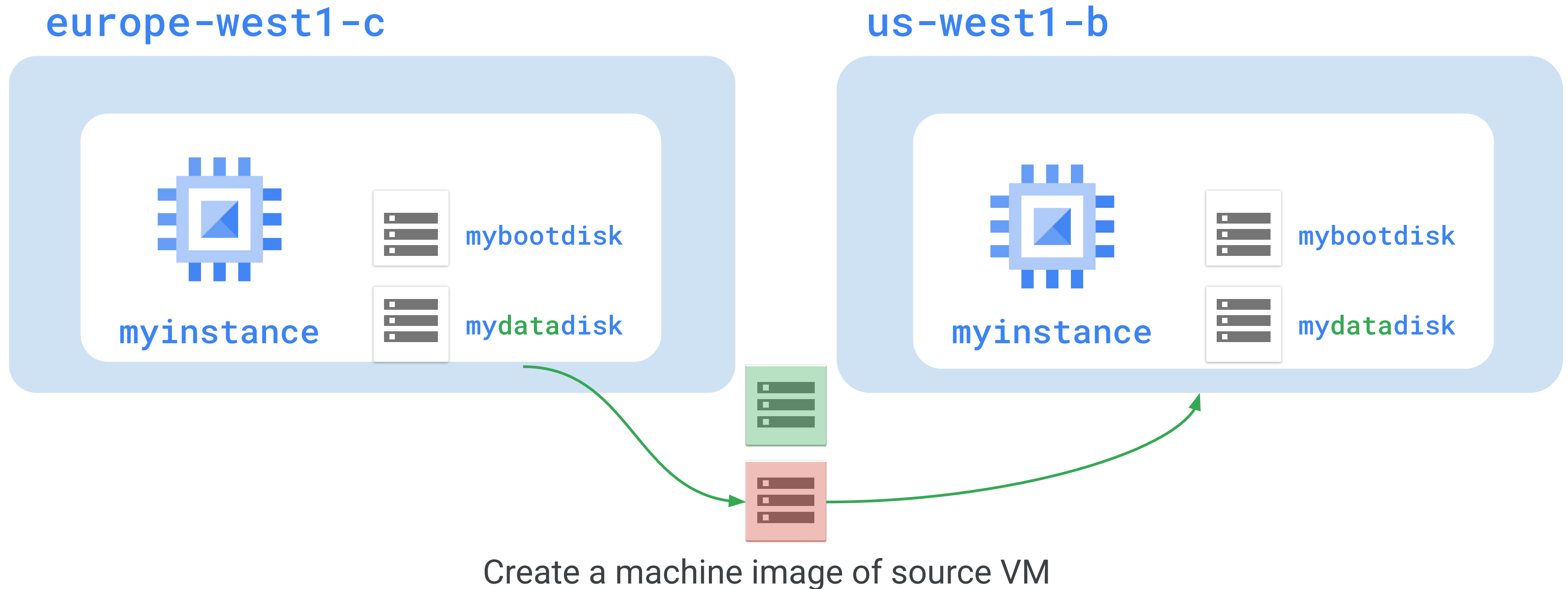
# Common Compute Engine Actions

# Metadata and scripts

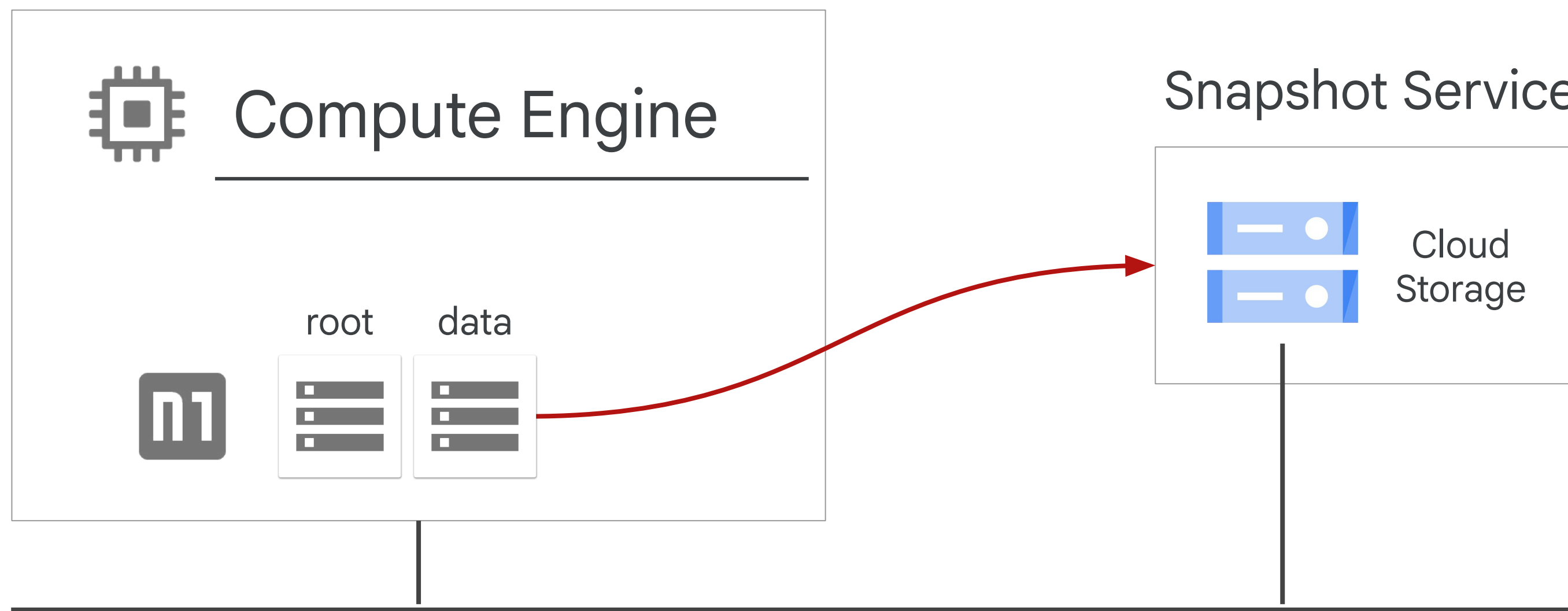




# Move an instance to a new zone or region



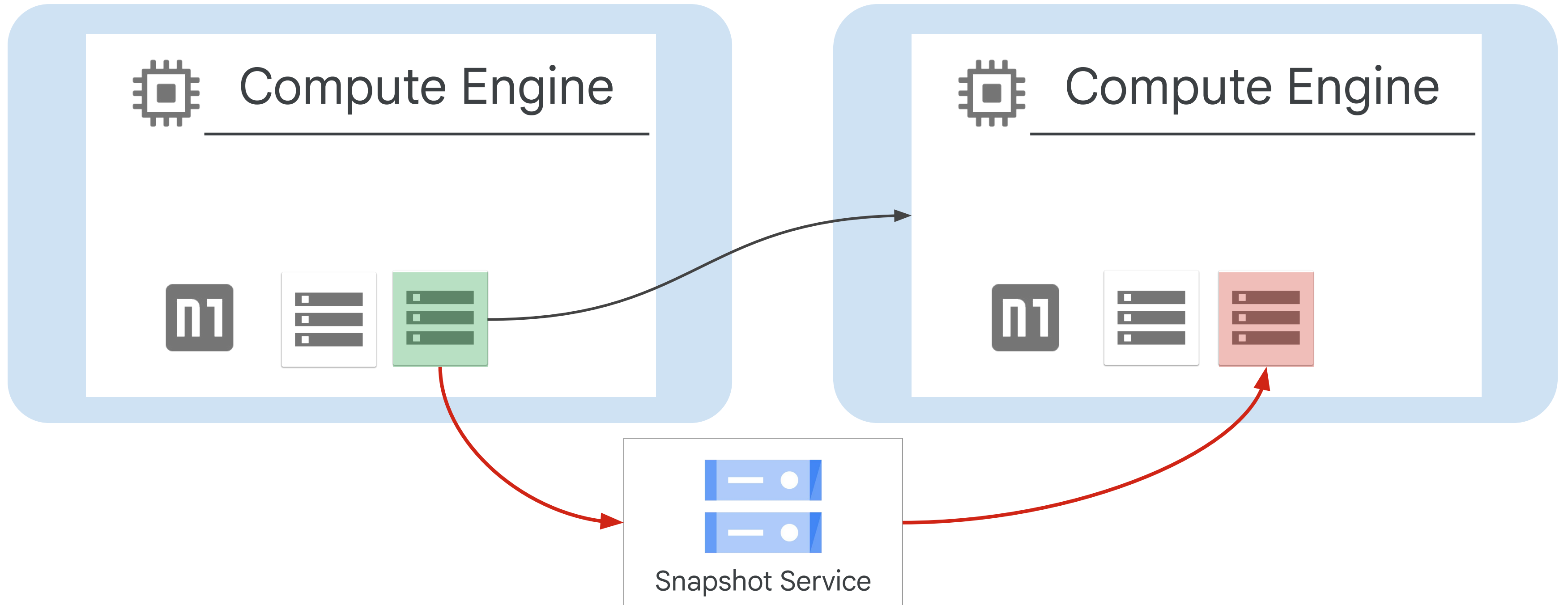
# Snapshot: Backup critical data



# Snapshot: Migrate data between zones

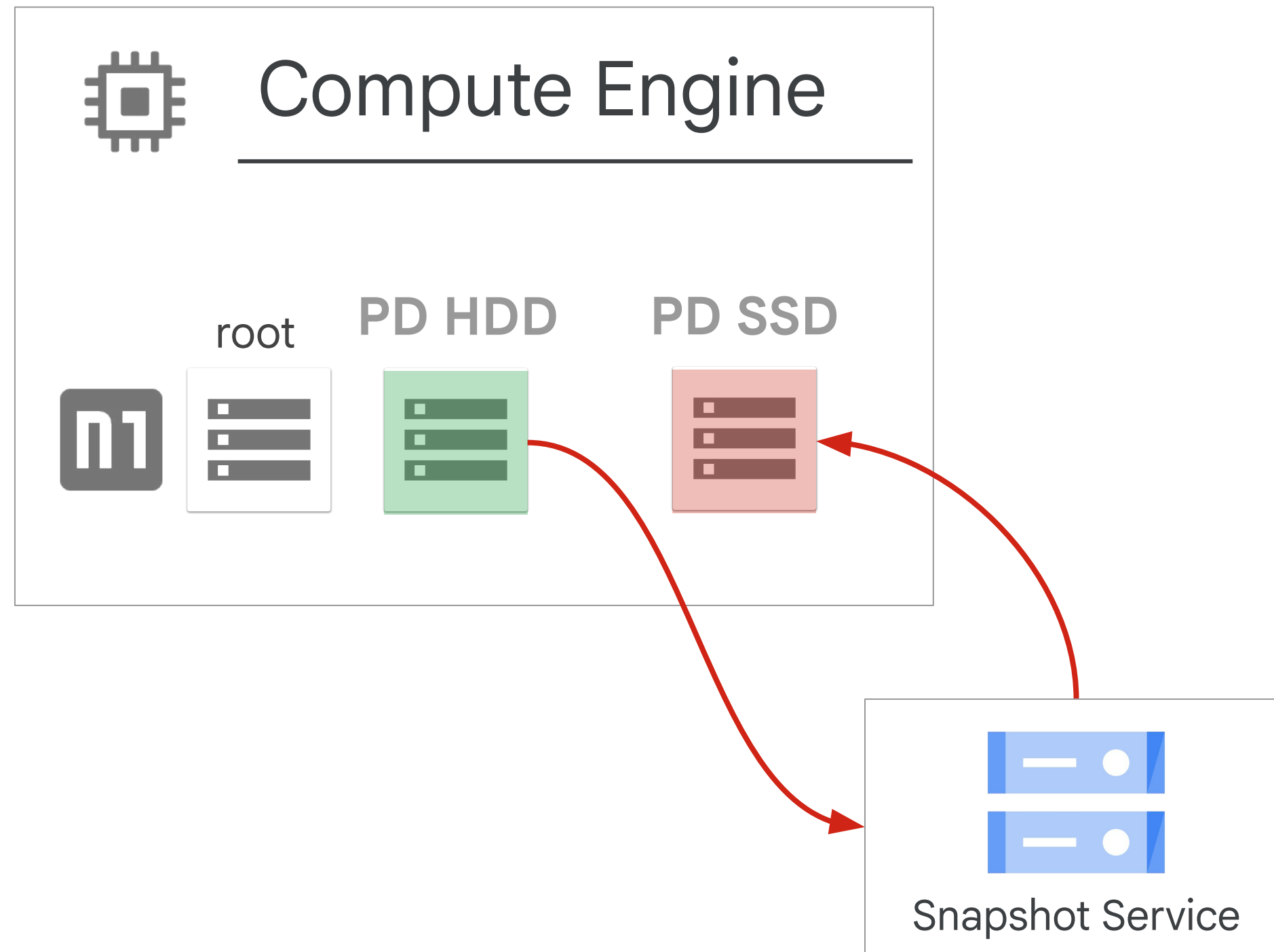
Zone 1

Zone 2





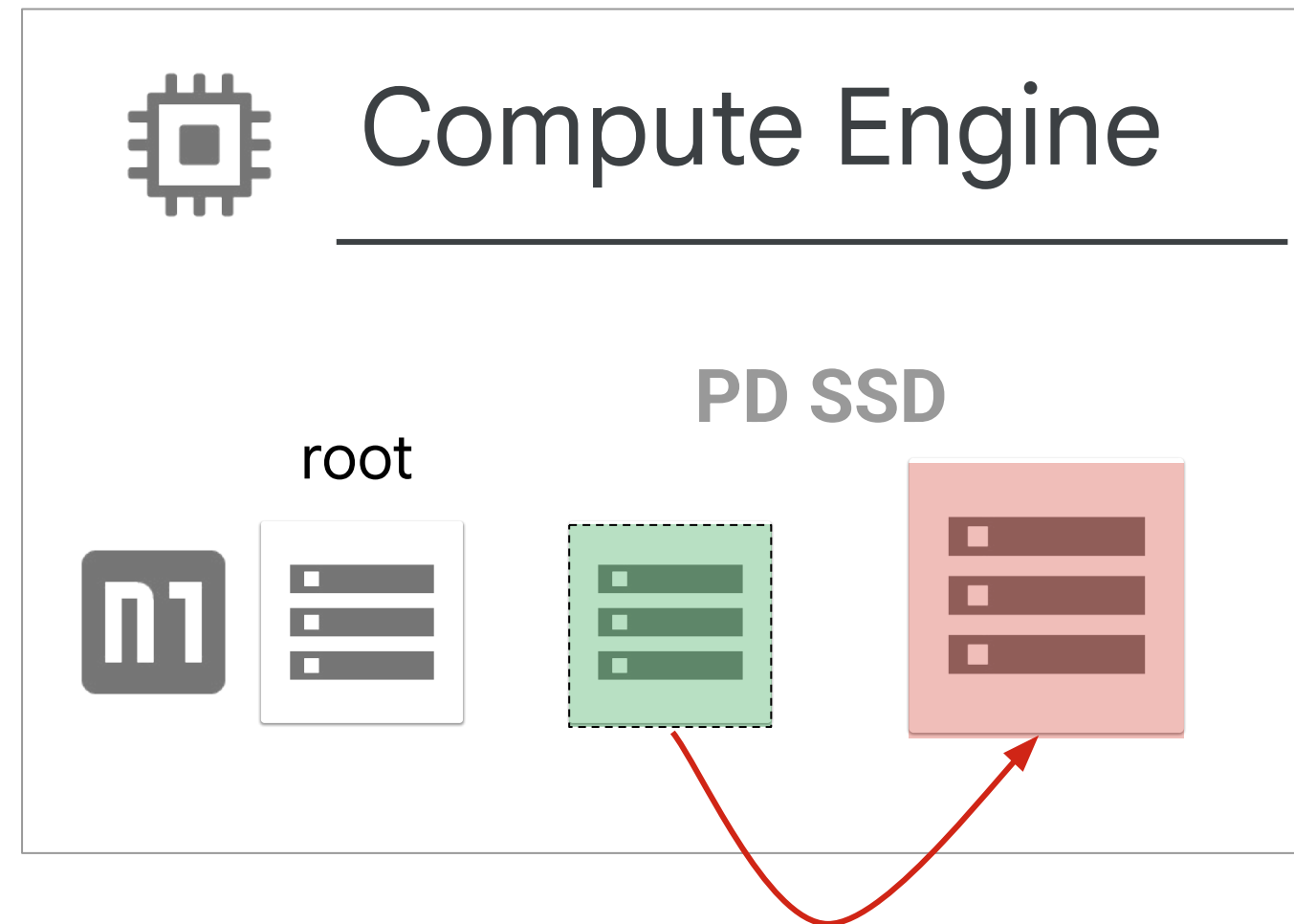
# Snapshot: Transfer to SSD to improve performance



# Persistent disk snapshots

- Snapshot is not available for local SSD.
- Creates an *incremental* backup to Cloud Storage.
  - Not visible in your buckets; managed by the snapshot service.
- Create scheduled snapshots.
  - Regularly and automatically back up your zonal and regional persistent disks.
- Snapshots can be restored to a new persistent disk.
  - New disk can be in another region or zone in the same project.
  - Basis of VM migration: "moving" a VM to a new zone.
    - Snapshot doesn't backup VM metadata, tags, etc.

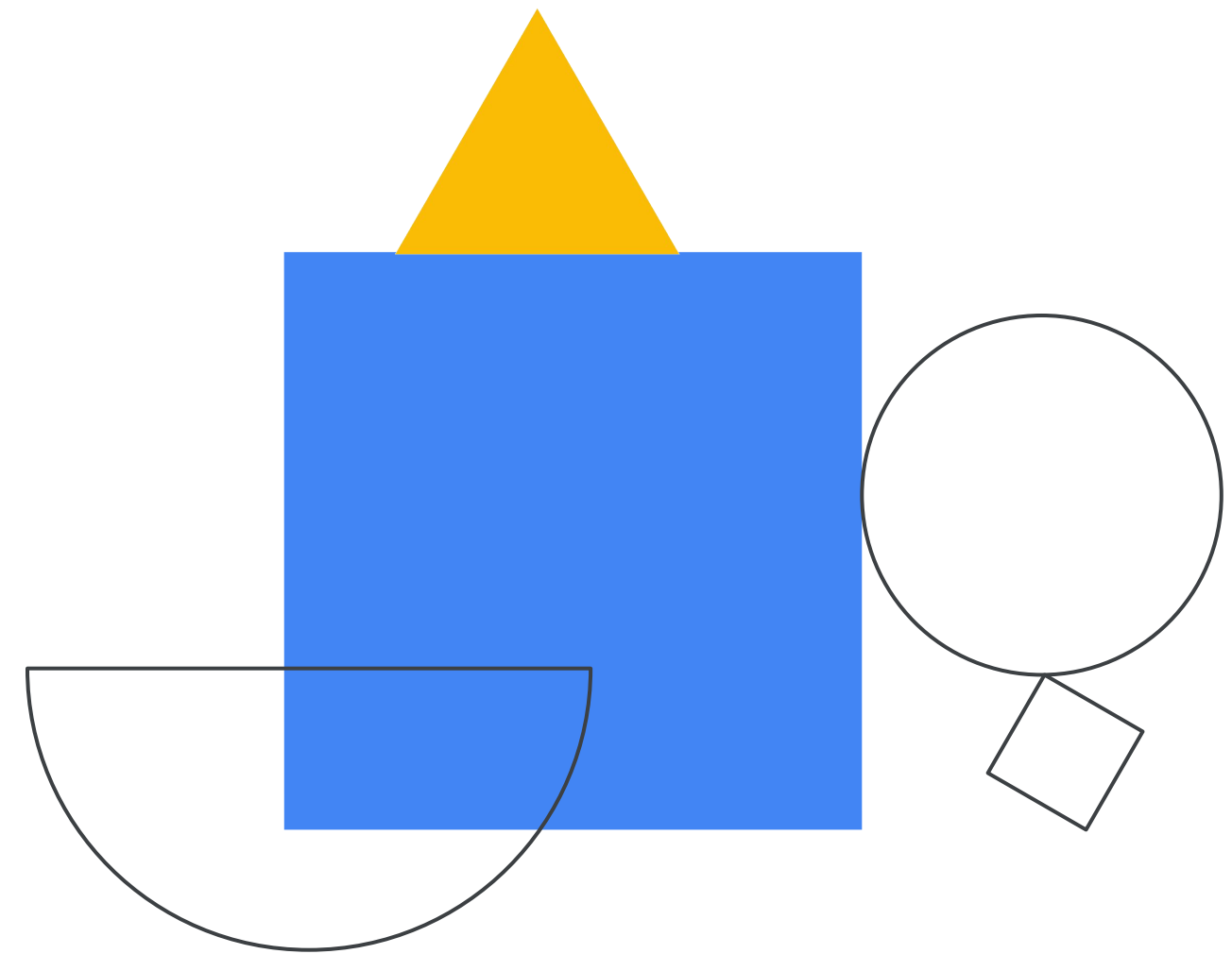
# Resize persistent disk



You can grow disks, but never shrink them!

# Lab Intro

Working with Virtual Machines



# Lab objectives

01

Customize an application server

02

Install and configure necessary software

03

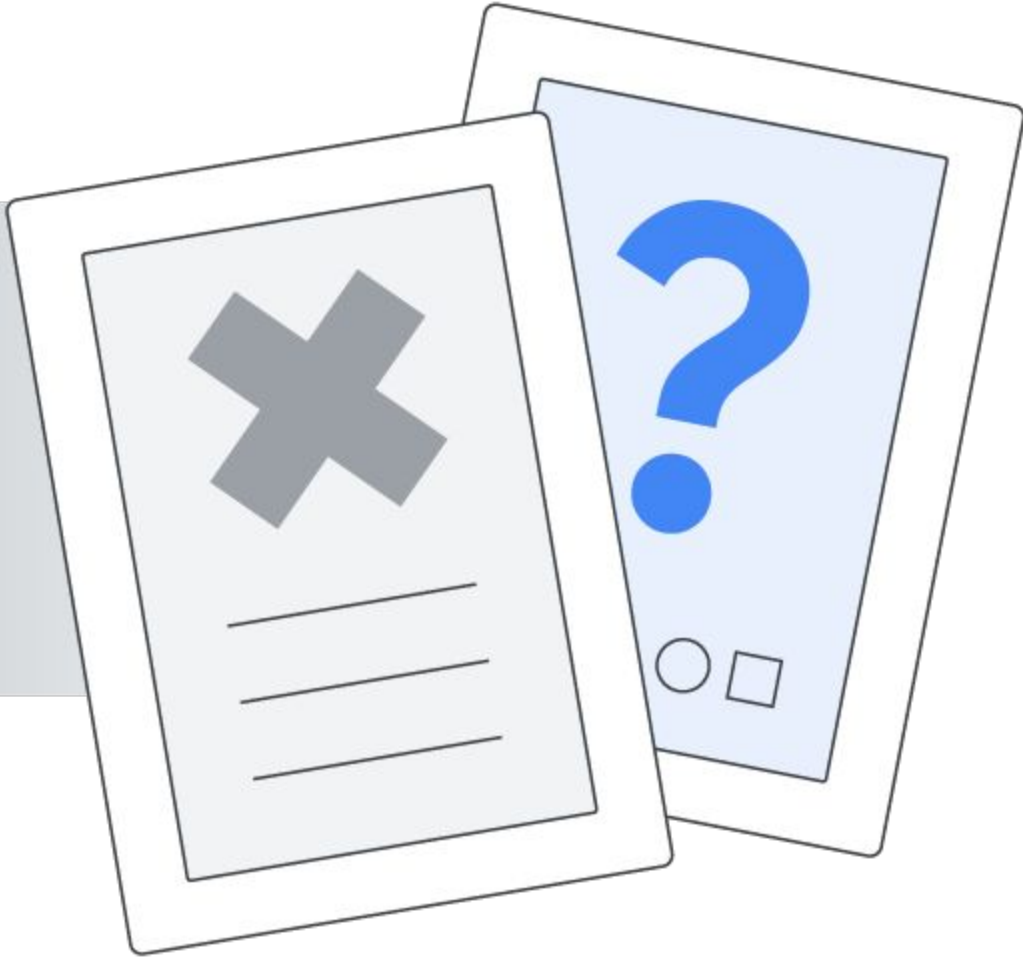
Configure network access

04

Schedule regular backups







# Quiz



# Question #1

## Question

Which statement is true of virtual machine instances in Compute Engine?

- A. A VM in Compute Engine always maps to a single hardware computer in a rack
- B. Compute Engine uses VMware to create virtual machine instances
- C. In Compute Engine, a VM is a networked service that simulates the features of a computer
- D. All Compute Engine VMs are single tenancy and do not share CPU hardware

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## Answer

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# Question #2

## Question

What are sustained use discounts?

- A. Purchase commitments for specific resources you know you will use
- B. Automatic discounts that you get for running specific Compute Engine resources for a significant portion of the billing month
- C. Discounts you receive by using preemptible VM instances
- D. Per-second billing that starts after a 1 minute minimum

# Question #2

## Answer

What are sustained use discounts?

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# Question #3

## Question

Which statement is true of persistent disks?

- A. Persistent disks are encrypted by default
- B. Persistent disks are physical hardware devices connected directly to VMs
- C. Persistent disks are always HDDs (magnetic spinning disks)
- D. Once created, a persistent disk cannot be resized

# Question #3

## Answer

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# Review: Virtual Machines

